Unit 3.01: Sequences, Series & Binomial Theorem

Q1. The first four terms of an arithmetic sequence are 5, 12, 19, and 26. What is the 20th term of this sequence?

[ ] A: 131
[x] B: 138
[ ] C: 140
[ ] D: 145

Solution: To find the 20th term of the arithmetic sequence, first identify the initial term and the common difference: 1. The first term is $a_1 = 5$. 2. The common difference d is found by subtracting consecutive terms: $d = 12 - 5 = 7$. The general formula for the n th term of an arithmetic sequence is: $a_n = a_1 + n - 1d$ Substitute $n = 20$, $a_1 = 5$, and $d = 7$: $a_{20} = 5 + 20 - 17 = 5 + 197 = 5 + 133 = 138$ Common misconceptions: - Using n instead of n - 1 yields $5 + 207 = 145$. - Calculating 20 times 7 = 140 overlooks the starting value $a_1 = 5$. - Using n - 2 steps yields $5 + 187 = 131$.

Q2. In an arithmetic sequence, the 3rd term is k + 7 , the 6th term is 3k - 1 , and the 11th term is 6k - 8 , where k is a real constant. What is the first term, a_1 , of this sequence?

[x] A: 6
[ ] B: 10
[ ] C: 19
[ ] D: 26

Solution: In an arithmetic sequence with first term a_1 and common difference d , the n th term is given by $a_n = a_1 + n - 1d$. Step 1: Express the common difference d in terms of k using the differences between the given terms: $a_6 - a_3 = 3d = 3k - 1 - k + 7 = 2k - 8$ implies $d = \frac{2k - 8}{3}$ $a_{11} - a_6 = 5d = 6k - 8 - 3k - 1 = 3k - 7$ implies $d = \frac{3k - 7}{5}$ Step 2: Equate the two expressions for d to solve for k : $\frac{2k - 8}{3} = \frac{3k - 7}{5}$ $52k - 8 = 33k - 7$ $10k - 40 = 9k - 21$ $k = 19$ Step 3: Determine d and a_3 : $d = \frac{219 - 8}{3} = \frac{30}{3} = 10$ $a_3 = 19 + 7 = 26$ Step 4: Find the first term a_1 : $a_1 = a_3 - 2d = 26 - 210 = 6$ Common traps: - Selecting 19 confuses the parameter k with the first term. - Selecting 10 stops at finding the common difference d . - Selecting 26 stops at the value of the 3rd term a_3 .

Q3. An arithmetic sequence a_n and a geometric sequence b_n of non-zero real numbers share the same first term, so $a_1 = b_1$. The sequences also satisfy $a_5 = b_2$ and $a_6 = b_3$, with $b_2 \neq b_1$. If the maximum value of the partial sum $S_N = \sum_{k=1}^{N} a_k$ of the arithmetic sequence is 153 , what is the sum of the infinite geometric series $\sum_{n=1}^{\infty} b_n$?

[x] A: 64
[ ] B: 96
[ ] C: 144
[ ] D: 192

Solution: Let $a_1 = b_1 = a$, let d be the common difference of a_n , and let r be the common ratio of b_n . 1. Express the given terms: $a_5 = a + 4d = b_2 = ar$ $a_6 = a + 5d = b_3 = ar^2$ Subtracting the first equation from the second gives: $d = a_6 - a_5 = ar^2 - ar = arr - 1$ Substitute d back into $a + 4d = ar$: $a + 4arr - 1 = ar$ Since the terms of the sequence are non-zero, $a \neq 0$. Dividing by a : $1 + 4rr - 1 = r$ implies $4r^2 - 5r + 1 = 0$ $4r - 1r - 1 = 0$ Since $b_2 \neq b_1$, $r \neq 1$. Therefore, $r = \frac{1}{4}$. 2. Determine d in terms of a : $d = arr - 1 = a\left(\frac{1}{4}\right)\left(-\frac{3}{4}\right) = -\frac{3}{16}a$ 3. Optimize the partial sum S_N : For S_N to achieve a finite maximum, the common difference must be negative, so $d < 0$ implies $a > 0$. The n -th term of the arithmetic sequence is: $a_n = a + n - 1\left(-\frac{3}{16}a\right) = \frac{a}{16}(19 - 3n)$ Evaluating the sign of a_n : - For $n \leq 6$, $19 - 3n > 0$ implies $a_n > 0$ - For $n \geq 7$, $19 - 3n < 0$ implies $a_n < 0$ Because terms are added until they become negative, S_N attains its unique maximum at $N = 6$. 4. Solve for a : $S_6 = \frac{6}{2}a_1 + a_6 = 3\left(a + \frac{a}{16}\right) = 3 \cdot \frac{17a}{16} = \frac{51a}{16}$ Given $S_6 = 153$: $\frac{51a}{16} = 153$ implies $a = 153 \cdot \frac{16}{51} = 3 \cdot 16 = 48$ 5. Sum of the infinite geometric series: Since $|r| = \frac{1}{4} < 1$, the infinite series converges to: $\sum_{n=1}^{\infty} b_n = \frac{b_1}{1 - r} = \frac{48}{1 - \frac{1}{4}} = \frac{48}{\frac{3}{4}} = 64$

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Q4. The bar chart below shows the values of the first four terms of an arithmetic sequence, a_1, a_2, a_3, and a_4, where a_1 = 6, a_2 = 11, a_3 = 16, and a_4 = 21. What is the value of the 8th term, a_8, of this sequence?

[ ] A: 36
[x] B: 41
[ ] C: 46
[ ] D: 48

Solution: 1. Find the common difference d by subtracting consecutive terms: d = a_2 - a_1 = 11 - 6 = 5 Check consistency with subsequent terms: 16 - 11 = 5 and 21 - 16 = 5. 2. Use the standard formula for the nth term of an arithmetic sequence: a_n = a_1 + n - 1d For the 8th term n = 8: a_8 = 6 + 8 - 15 = 6 + 75 = 6 + 35 = 41 Alternatively, continuing term-by-term: a_5 = 26, a_6 = 31, a_7 = 36, a_8 = 41. Common Errors: - Choosing 36 corresponds to finding the 7th term a_7 due to an off-by-one error. - Choosing 46 results from using a_1 + 8d instead of a_1 + 8 - 1d. - Choosing 48 results from incorrectly treating the sequence as direct multiplication, 6 times 8.

Q5. The first three terms of a geometric sequence of positive real numbers are k - 1, k + 2, and 3k, in that order. What is the value of the 6th term of this sequence?

[ ] A: 24
[ ] B: 48
[x] C: 96
[ ] D: 192

Solution: Step 1: Set up the ratio property for consecutive terms of a geometric sequence. Since the sequence is geometric, the common ratio r satisfies: r = frac{a_2}{a_1} = frac{a_3}{a_2} implies frac{k + 2}{k - 1} = frac{3k}{k + 2} Step 2: Solve the quadratic equation for k. Cross-multiplying gives: k + 2^2 = 3kk - 1 k^2 + 4k + 4 = 3k^2 - 3k 2k^2 - 7k - 4 = 0 Factoring the quadratic: 2k + 1k - 4 = 0 Thus, k = -frac{1}{2} or k = 4. Step 3: Discard the extraneous solution. The problem states that all terms in the sequence are positive real numbers. If k = -frac{1}{2}, the first term is k - 1 = -frac{3}{2} < 0, which contradicts the condition. Therefore, k = 4. Step 4: Find the first term and common ratio. - First term: a_1 = 4 - 1 = 3 - Second term: a_2 = 4 + 2 = 6 - Common ratio: r = frac{6}{3} = 2 Step 5: Compute the 6th term. Using the formula for the nth term of a geometric sequence, a_n = a_1 r^{n-1}: a_6 = 3 cdot 2^{6 - 1} = 3 cdot 2^5 = 3 cdot 32 = 96 Misconception Traps: - Choice A 24 is the 4th term 3 cdot 2^3. - Choice B 48 is the 5th term 3 cdot 2^4, resulting from an off-by-one exponent error. - Choice D 192 incorrectly uses a_n = a_1 r^n giving 3 cdot 2^6 = 192.

Q6. An infinite geometric sequence of positive real numbers has first term a_1 and common ratio r in 0, 1. Let S_1 = sum_{n=1}^infty a_n, S_2 = sum_{n=1}^infty a_n^2, and S_3 = sum_{n=1}^infty a_n^3. If S_1 = 4 and S_2 = 8, what is the value of S_3?

[x] A: frac{256}{13}
[ ] B: frac{128}{13}
[ ] C: frac{64}{7}
[ ] D: 16

Solution: For an infinite geometric sequence with first term a_1 > 0 and ratio r in 0, 1, the terms are a_n = a_1 r^{n-1}. 1. The sum of the original sequence is: S_1 = frac{a_1}{1 - r} = 4 implies a_1 = 41 - r 2. The terms of the squared sequence are a_n^2 = a_1^2 r^2^{n-1}, which forms a geometric sequence with first term a_1^2 and common ratio r^2. Its sum is: S_2 = frac{a_1^2}{1 - r^2} = 8 3. Relate S_2 and S_1 using the algebraic identity 1 - r^2 = 1 - r1 + r: frac{S_2}{S_1^2} = frac{frac{a_1^2}{1 - r^2}}{leftfrac{a_1}{1 - r}right^2} = frac{1 - r^2}{1 - r^2} = frac{1 - r}{1 + r} Substituting the known numerical values S_1 = 4 and S_2 = 8: frac{8}{4^2} = frac{8}{16} = frac{1}{2} implies frac{1 - r}{1 + r} = frac{1}{2} Cross-multiplying gives: 21 - r = 1 + r implies 2 - 2r = 1 + r implies 3r = 1 implies r = frac{1}{3} 4. Find the first term a_1: a_1 = 4left1 - frac{1}{3}right = 4 cdot frac{2}{3} = frac{8}{3} 5. The cubed sequence has terms a_n^3 = a_1^3 r^3^{n-1}, with first term a_1^3 and ratio r^3. S_3 = frac{a_1^3}{1 - r^3} = frac{leftfrac{8}{3}right^3}{1 - leftfrac{1}{3}right^3} = frac{frac{512}{27}}{1 - frac{1}{27}} = frac{frac{512}{27}}{frac{26}{27}} = frac{512}{26} = frac{256}{13} Common misconception: Dividing S_2 directly by S_1 to get frac{a_1}{1+r} = 2 and

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Q7. The first four terms of an arithmetic sequence are shown below. 4, 11, 18, 25, dots What is the 31st term of the sequence?

[ ] A: 210
[x] B: 214
[ ] C: 217
[ ] D: 221

Solution: To find the 31st term of an arithmetic sequence, use the formula for the nth term: a_n = a_1 + n - 1d 1. Identify the first term a_1: a_1 = 4 2. Find the common difference d by subtracting consecutive terms: d = 11 - 4 = 7 3. Substitute n = 31, a_1 = 4, and d = 7 into the formula: a_{31} = 4 + 31 - 17 = 4 + 307 = 4 + 210 = 214 - A common error is using n instead of n - 1, which yields 4 + 317 = 221. - Another common error is computing 31 times 7 = 217 or forgetting to add the initial term a_1 = 4, giving 210.

Q8. The second term of a geometric sequence of positive real numbers is k , and the fifth term is k^2 - 20 . If the common ratio of the sequence is 2 , what is the sum of the first 6 terms of the sequence?

[ ] A: 155
[ ] B: 160
[x] C: 315
[ ] D: 630

Solution: Step 1: Express the terms in terms of the first term a_1 and common ratio r = 2 . The n -th term of a geometric sequence is given by a_n = a_1 r^{n-1} . For the second term: a_2 = a_12^{2-1} = 2a_1 = k implies a_1 = frac{k}{2} For the fifth term: a_5 = a_12^{5-1} = 16a_1 Substituting a_1 = frac{k}{2} gives: a_5 = 16leftfrac{k}{2}right = 8k Step 2: Solve for the parameter k . We are given a_5 = k^2 - 20 , so: k^2 - 20 = 8k implies k^2 - 8k - 20 = 0 Factoring the quadratic equation: k - 10k + 2 = 0 Since the sequence consists of positive real numbers, k > 0 , so k = 10 . Step 3: Determine a_1 and compute the sum S_6 . The first term is a_1 = frac{10}{2} = 5 . The sum of the first n terms of a geometric sequence is given by S_n = frac{a_1r^n - 1}{r - 1} . For n = 6 : S_6 = frac{52^6 - 1}{2 - 1} = 564 - 1 = 563 = 315 Common traps: - Confusing the sum with the 6th term gives a_6 = 52^5 = 160 . - Summing only 5 terms gives S_5 = 531 = 155 . - Forgetting to divide k by 2 to find a_1 mistaking a_2 for a_1 gives 1063 = 630 .